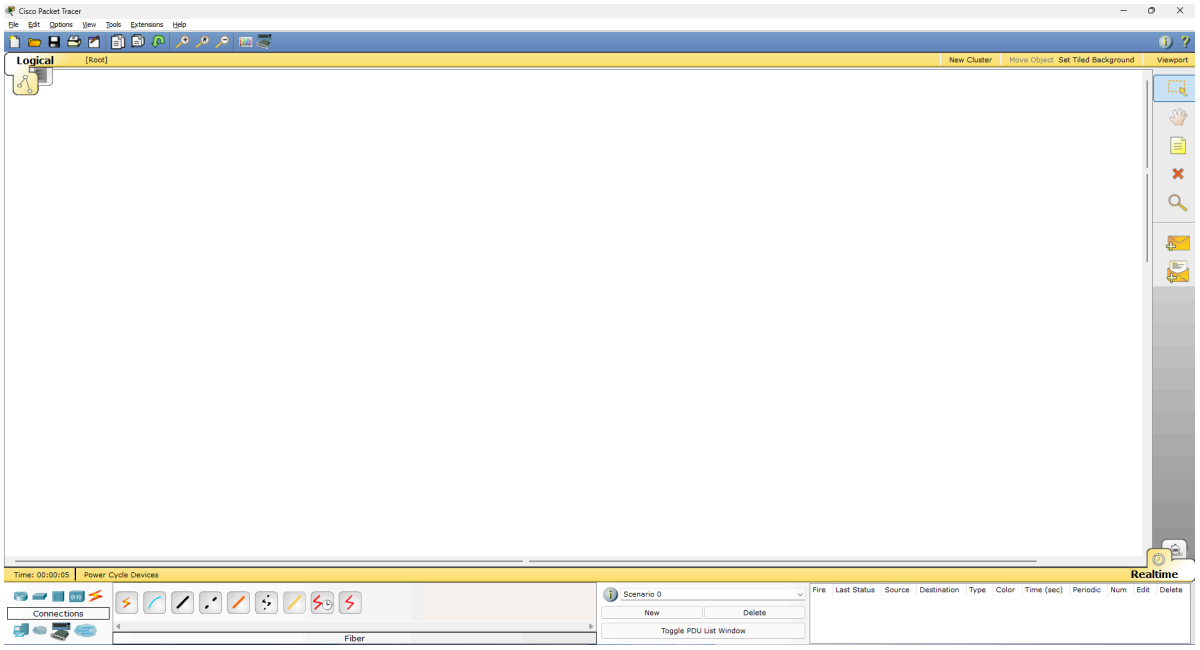
PRACTICALS

Computer Networks Laboratory

Practical 6.

Aim: To implement the topology having multiple routers (at least 3 routers with Class A, Class B and Class C IP address) to reach the destination following static routing in packet tracer simulator.

Step 1: Open Cisco packet tracer 5.1 as shown in **FIG 1**

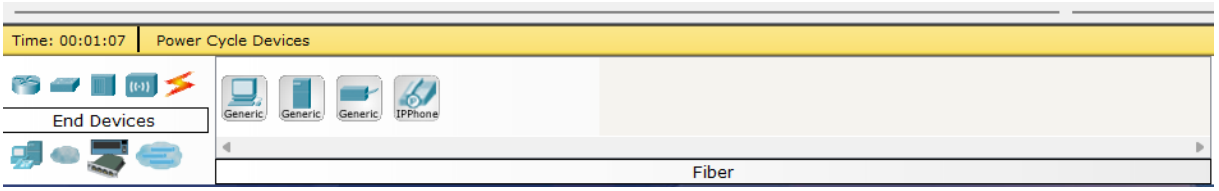


[FIG 1]

Step 2: Select device to make LAN (Local Area Network) Network form the bottom Tool bar as shown in **FIG 2**

There are multiple devices used in this process are follows as:

1. Network Switch
2. Router
3. Hub
4. Wireless device
5. Cables/wires (for connection)
6. End devices (PC, Laptop, Server etc)
7. WAN Enulation
8. Custom made device
9. Multi User connection



[FIG 2]

Step 3: Select PCs, Network Switch, & connecting wires/Cables Shown in **FIG 3**

1. 6 - PCs (3 for each Network)
2. 2 - Network Switch (1 For each Network)
3. 11 - Connecting wires/ cables (5 For Each Network and 1 common For both Network)
4. 2 - Server (1 For Each Network)
5. 2 - Router (1 For Each Network)



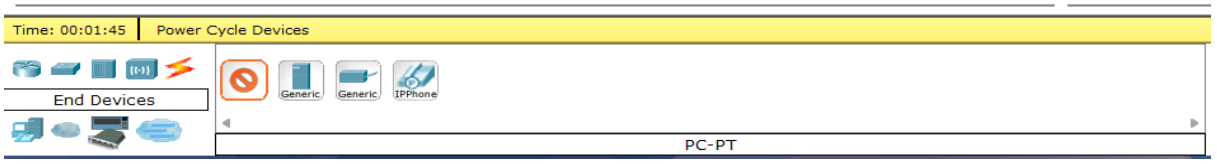


FIG 3

Step 4: Creating Network Using Required devices as shown in **FIG 4**

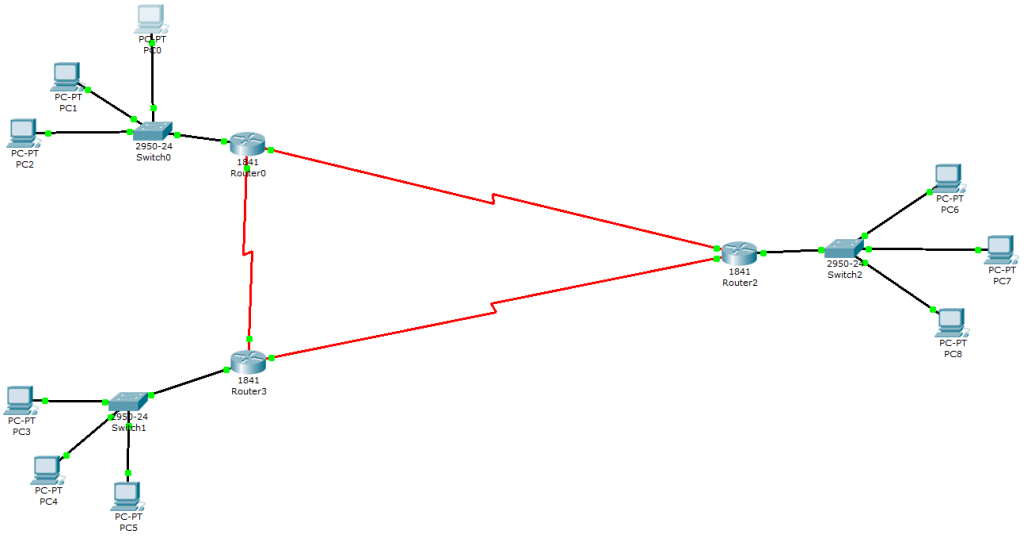


FIG 4

Step 5: Configuring The Setup with Network (DHCP method) Using Default Gateway, Subnet Mask, Ip Address as shown in **FIG 5** and **FIG 6**

Note: This Process Is Only For One Network Do Same Process For All The Networks

1. Go To PC
2. Desktop Tab
3. Click on Ip Configuration

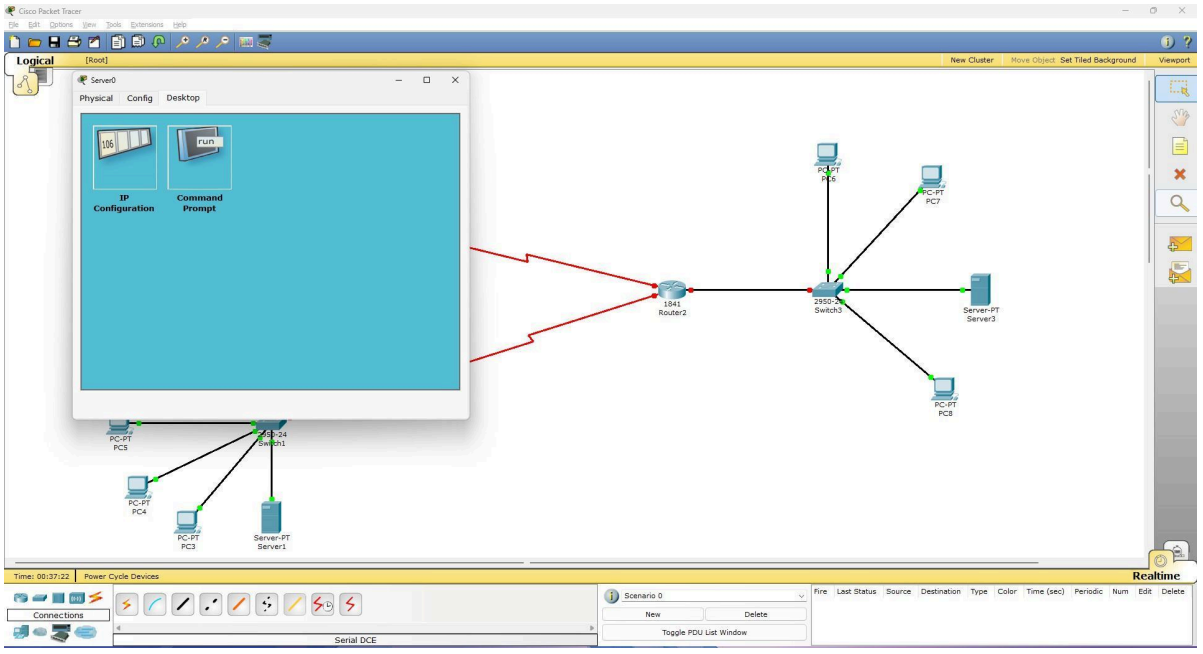


FIG 5

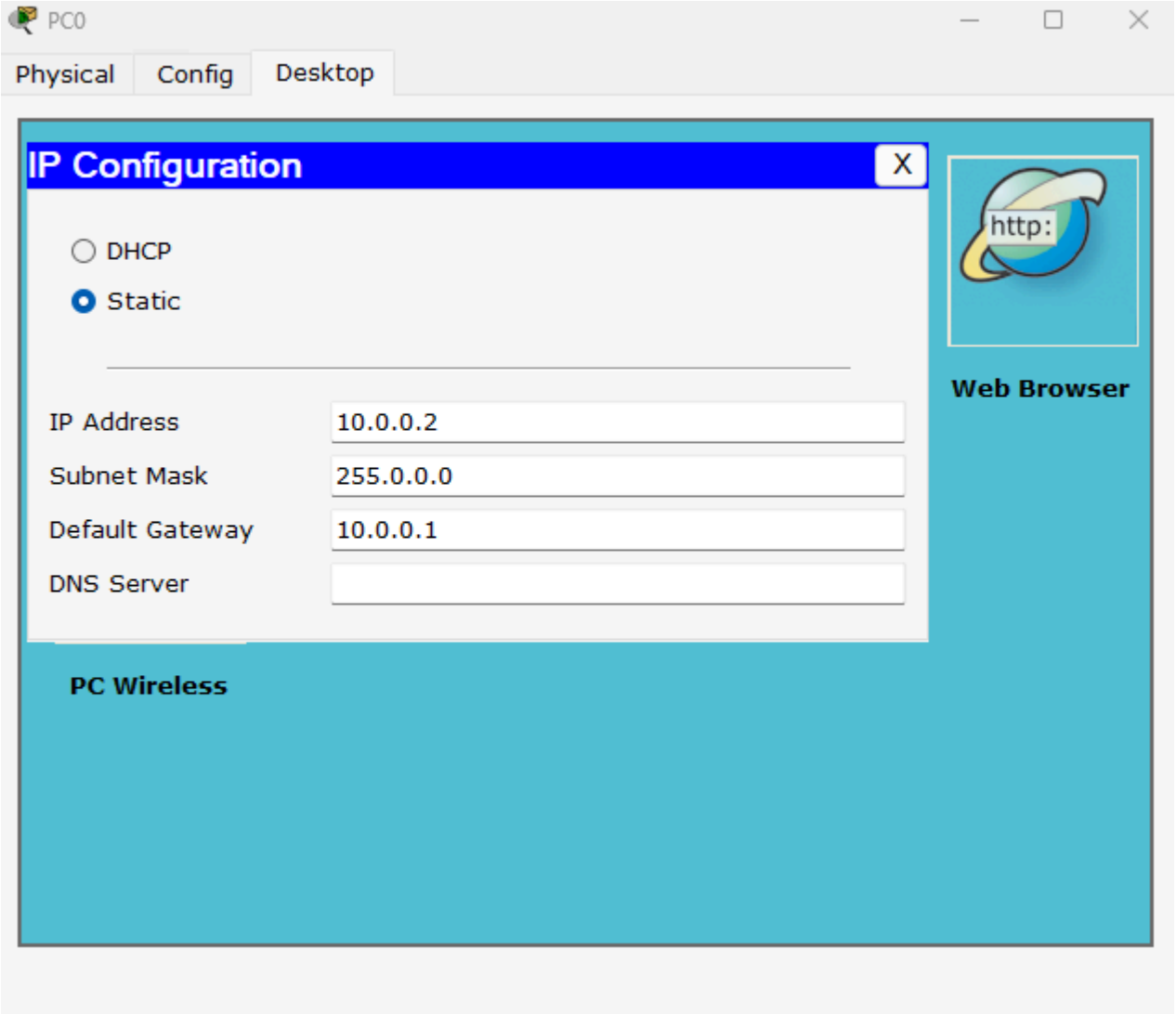


FIG 6

Step 7: Now to connect all the router we need **WIC-2T**

1. Select Router
2. Under Physical Tab you will find out **WIC-2T** Second last option
3. Turn off the switch (there is toggle button Exactly under Zoom Out button)
4. Drag and drop WIC-2T into a space between Toggle switch and Few other ports as Shown in **FIG 8**
5. Turn on the Toggle Switch
6. Do same process for all other routers

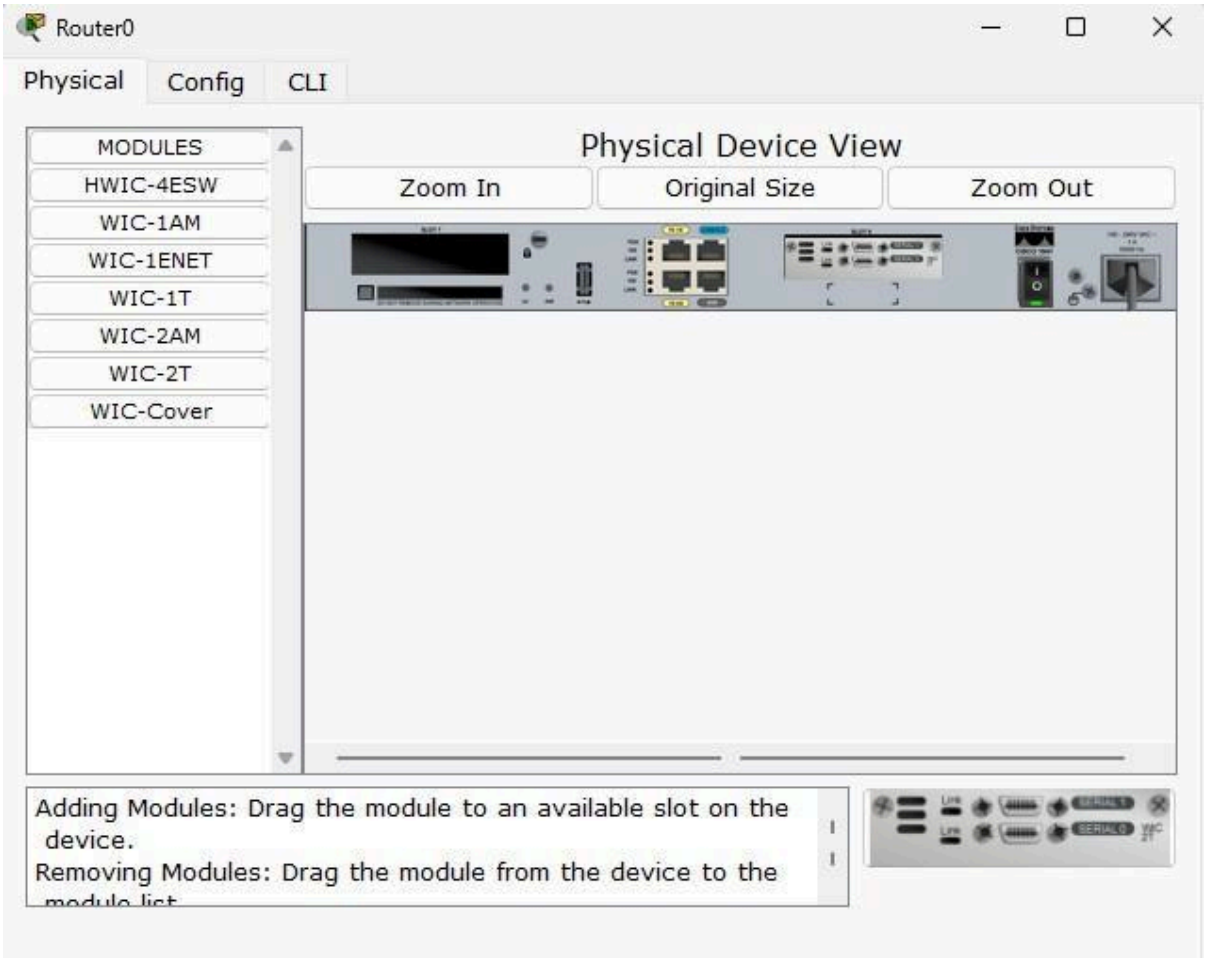

Router0
Physical
Config
CLI
MODULES
HWIC-4ESW
WIC-1AM
WIC-1ENET
WIC-1T
WIC-2AM
WIC-2T
WIC-Cover
Physical Device View
Zoom In
Original Size
Zoom Out
Adding Modules: Drag the module to an available slot on the device.
Removing Modules: Drag the module from the device to the

FIG 8

Step 8: Now Configuring Routers with the following Commands:
We have to run this commands in CLI of router, To open CLI just click on router and go to CLI and Run the Following Commands

Router1 Configuration (Network 1 side):
Open CLI of Router 1

```
enable
configure terminal
hostname Router0

interface fastEthernet0/0
 ip address 10.0.0.1 255.0.0.0
 no shutdown
 exit

interface serial0/0/0
 ip address 192.168.10.1 255.255.255.252
 clock rate 64000
 no shutdown
 exit

interface serial0/0/1
 ip address 192.168.20.1 255.255.255.252
 clock rate 64000
 no shutdown
 exit

ip route 172.16.0.0 255.255.0.0 192.168.10.2
ip route 192.168.1.0 255.255.255.0 192.168.20.2

end
write memory
```

Router2 Configuration (Network 2 side):
Open CLI of Router 2

```
enable
configure terminal
hostname Router1
```

```
interface fastEthernet0/0
 ip address 172.16.0.1 255.255.0.0
 no shutdown
 exit

interface serial0/0/0
 ip address 192.168.10.2 255.255.255.252
 clock rate 64000
 no shutdown
 exit

interface serial0/0/1
 ip address 192.168.30.1 255.255.255.252
 clock rate 64000
 no shutdown
 exit

ip route 10.0.0.0 255.0.0.0 192.168.10.1
ip route 192.168.1.0 255.255.255.0 192.168.30.2

end
write memory
```

Router3 Configuration (Network 3 side):
Open CLI of Router 3

```
enable
configure
terminal
hostname
Router2

interface
fastEthernet0/
0
 ip address
192.168.1.1
255.255.255.0
 no shutdown
 exit
```

```
interface
serial0/0/0
 ip address
192.168.30.2
255.255.255.25
2
 clock rate
64000
 no shutdown
 exit

interface
serial0/0/1
 ip address
192.168.20.2
255.255.255.25
2
 clock rate
64000
 no shutdown
 exit

ip route
10.0.0.0
255.0.0.0
192.168.20.1
ip route
172.16.0.0
255.255.0.0
192.168.30.1

end
write memory
```

Step 9: After compilation of step 6 in the bottom Tool bar's right side we can check the status of sending packets that it's successfully Send or Failed Shown in **FIG 9.1** For Success and **FIG 9.2** For Failed

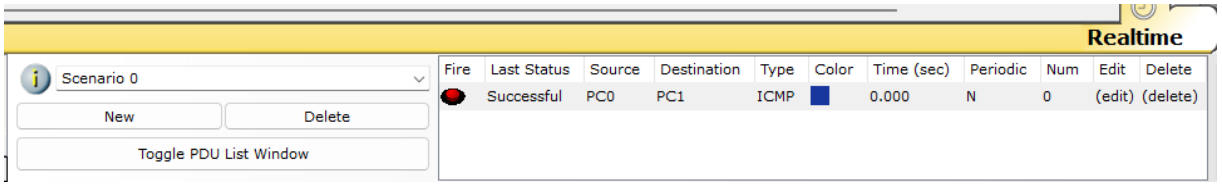


FIG 9.1

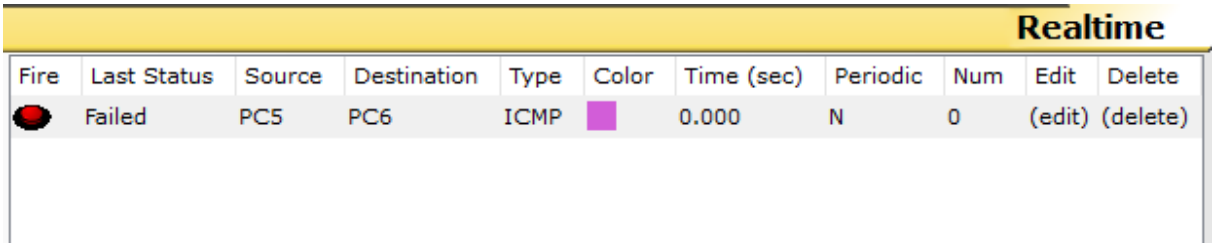


FIG 9.2